

Supervisory Controller Test Plan

- ☐ Able to receive and print all CAN messages
 - ☐ Receives all **Elevator Controller** messages (ID: 0x101)
 - ☐ Receives data: 0x00
 - ☒ ~~Receives data: 0x01~~
 - ☒ ~~Receives data: 0x02~~
 - ☒ ~~Receives data: 0x03~~
 - ☒ ~~Receives data: 0x05~~
 - ☒ ~~Receives data: 0x06~~
 - ☒ ~~Receives data: 0x07~~
 - ☒ Receives all **Car Controller** messages (ID: 0x200)
 - ☒ ~~Receives data: 0x01~~
 - ☒ ~~Receives data: 0x02~~
 - ☒ ~~Receives data: 0x03~~
 - ☒ ~~Receives Floor Controller 1 request (ID: 0x201)~~
 - ☒ ~~Receives Floor Controller 2 request (ID: 0x202)~~
 - ☒ ~~Receives Floor Controller 3 request (ID: 0x203)~~
- ☒ Able to send all CAN messages
 - ☒ Sends all **Elevator Controller** messages (ID: 0x101)
 - ☒ ~~Sends go to floor 1 (data: 0x05)~~
 - ☒ ~~Sends go to floor 2 (data: 0x06)~~
 - ☒ ~~Sends go to floor 3 (data: 0x07)~~
- ☒ Able to receive CAN message and send corresponding CAN message
 - ☒ Receives request from **Car Controller** (ID: 0x200)
 - ☒ ~~Receives: ID: 0x200 data: 0x01 Sends: 0x05~~
 - ☒ ~~Receives: ID: 0x200 data: 0x02 Sends: 0x06~~
 - ☒ ~~Receives: ID: 0x200 data: 0x03 Sends: 0x07~~
 - ☒ Receives request from **Floor Controller 1** (ID: 0x201)
 - ☒ ~~Receives: ID: 0x201 data: 0x01 Sends: 0x05~~
 - ☒ Receives request from **Floor Controller 2** (ID: 0x202)
 - ☒ ~~Receives: ID: 0x202 data: 0x01 Sends: 0x06~~
 - ☒ Receives request from **Floor Controller 3** (ID: 0x203)
 - ☒ ~~Receives: ID: 0x203 data: 0x01 Sends: 0x07~~

Above is a test plan for our Supervisory Controller. The goal of this text plan is to ensure that the Supervisory Controller is receiving all CAN messages and sending all CAN messages. Once this is confirmed a queue can be added to the Supervisory Controller.

This text plan may require some help from the STM32s. It is difficult for us to controller exactly what the Elevator Controller is sending on the CAN bus because the Arduino's code is given and should work as intended. To work around this an STM32 can be programmed with the Elevator Controller's ID (0x101) and send the data that the Elevator Controller will send. All the other messages can be tested using the STM32s as well because the message would typically be sent from the STM32 normally.

It is also encouraged that an STM32 is printing CAN message while working through the test plan. This would act as a double check ensuring that the Supervisory Controller is working as expected.

CAN ID (Hex)	Transmitter	Recipient(s)	DLC	Examples								Message value	
				Byte 0								Binary	Hex
				7	6	5	4	3	2	1	0		
0x100	SC	EC	1	0	0	0	0	0	1	1	1	b00000101	0x05
0x100	SC	EC	1	0	0	0	0	0	1	2		b00000110	0x06
0x100	SC	EC	1	0	0	0	0	0	1	3		b00000111	0x07
0x101	EC	ALL	1	0	0	0	0	0	1	1		b00000101	0x05
0x101	EC	ALL	1	0	0	0	0	0	1	2		b00000110	0x06
0x101	EC	ALL	1	0	0	0	0	0	1	3		b00000111	0x07
0x200	CC	SC	1	0	0	0	0	0	1	1		b00000101	0x05
0x200	CC	SC	1	0	0	0	0	0	1	2		b00000110	0x06
0x200	CC	SC	1	0	0	0	0	0	1	3		b00000111	0x07
0x200	CC	SC	1	0	0	0	0	0	0	1		b00000001	0x01
0x200	CC	SC	1	0	0	0	0	0	0	2		b00000010	0x02
0x200	CC	SC	1	0	0	0	0	0	0	3		b00000011	0x03
0x201	F1	SC	1	0	0	0	0	0	0	0	1	b00000001	0x01
0x202	F2	SC	1	0	0	0	0	0	0	0	1	b00000001	0x01
0x203	F3	SC	1	0	0	0	0	0	0	0	1	b00000001	0x01

In order to create a reasonable test environment I think that we should branch off of the code that we are working on right now. This will allow use to version control while going through the test plan without override our current code. This will also allow the current code to be referenced when necessary.